

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech.)

Sem - II University Examination, May- 2011

M.Tech. (CAD/CAM)

ME – 718 Computational Fluid Dynamics

Date: 20.05.2011, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

Q1 (a) State and derive the relation for Reynolds's Transport theorem. Derive a Continuity equation by using Reynolds's Transport theorem. **10**

(b) List down the commercially available CFD packages available in the market. **02**

OR

Q.1 (a) Discuss in detail the three basic methods of problem solving in engineering. Also discuss their relative merits and demerits **05**

(b) Explain the various components of Numerical Solution and errors in CFD tools. **05**

(c) Explain what is meant by truncation error. **02**

Q.2 (a) Explain the mathematical behavior of partial differential equations. Find out whether the following differential equations are elliptic, parabolic or hyperbolic: Laplace equation given by

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

(b) Enlist some important area of application of CFD. **02**

Q.3 (a) Explain the various Discretization techniques. **06**

(b) Explain structured grid and Unstructured grid **05**

OR

Q.3 (a) Explain the boundary fitted co-ordinate system **03**

(b) Explain the Various Convection Schemes (First order Upwind Scheme, Central difference Scheme, Quick Scheme). **08**

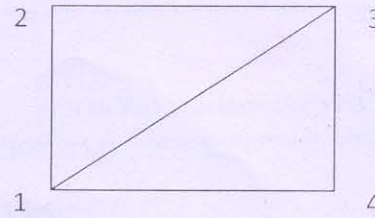
P.T.O

SECTION - II

Q4 (a) Explain the various steps used in FEM. 06

(b) Find the temperature at point P by FEM 06

Point	X-coordinate	Y-Coordinate
1	0	0
2	0	1
3	1	1
4	1	0
P	MM / 20	DD / 35



$$T1 = DD, T2 = 2 * DD,$$

$$T3 = \sqrt{YYYY}, T4 = MM$$

Where MM is your Birth month.

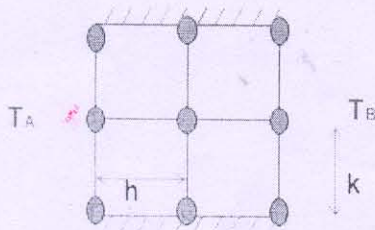
Q5 (a) Explain the Explicit and Implicit Approach for transient problem with its advantages and limitations. 04

(b) Using Taylor's series derive second order central-difference for $\frac{\partial u}{\partial y}$. 04

(c) Explain the polynomial approach to find a second order accurate finite difference at the boundary. 04

OR

Q5 Calculate the temperature at described point on the plate surface, the plate of uniform thickness, homogeneous in nature, subjected to two different temperature at two ends $T_A = 26^\circ\text{C}$ & $T_B = 44.4522^\circ\text{C}$ and other two ends are insulated, so, insulated so involves 2D heat transfer with governing equation. 12



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Q6 (a) What are Eigen Value problems? What are Eigen values & Eigen vectors Compute highest Eigen value and corresponding Eigen vectors for the system below 06

$$3x - y = 0, -2x + 4y - 3z = 0, -y + z = 0$$

(b) Explain "Trapezoidal Rule" and "Simson's 1/3rd rule". 05

OR

Q6 (a) Explain how hyper mesh will works 05

(b) Explain the CFL stability criteria. 06